

- 1) **Stakeholder:** Small Businesses and Financial Institutions.
- 2) **Interest/value:** For financial institutions, what matters most is regulatory compliance, threat mitigation, and dispute precision. They require precise classification that rapidly elevates actionable unauthorized fraud without triggering false positives that burden investigation pipelines or breach statutory resolution timelines. For small businesses, priorities center on operational fairness, revenue stability, and reputational integrity. They need reliable differentiation between severe financial misconduct and routine shipping delays so that urgent consumer phrasing does not cause unwarranted account freezes, chargeback penalties, or automated dispute escalation. Both groups depend on a calibrated, transparent system that curbs exploitation while safeguarding business continuity.
- 3)
 - a) **Potential harms:** Misclassification of complaints remains a potential harm. If our model misclassifies a fraud complaint, thereby reducing its severity, this could cause lawsuits and financial loss to businesses and financial institutions.
 - b) **Potential benefits:** In today's economy, businesses are often "dragged on social media" when they fail to address customer complaints promptly. With our model, which classifies complaints with about 92% accuracy, businesses and financial institutions can finally clear their complaint backlogs. They can classify and address issues swiftly and stay on top of future complaints.
- 4) **Financial Institutions:** Deliberative Workshops (action catalogue).
Small Businesses: Focus groups
- 5) **Financial Institutions' Engagement method:** Deliberative workshops are structured, in-depth engagement forums designed to convene non-technical decision-makers, sector specialists, and technical practitioners to dissect complex socio-technical mechanisms, weigh trade-offs, and co-develop governance boundaries. This method is particularly well-suited for financial institutions—such as retail banks, credit card networks, and payment processors—because integrating our language model (LM) into complaint triage involves navigating strict regulatory mandates, legal liability boundaries, and systemic operational friction. Conventional, one-directional engagement mechanisms such as surveys or focus groups fail in this context; they gather

superficial sentiment rather than addressing the precise legal constraints of statutory dispute-resolution windows and dispute substantiation rules.

A deliberative workshop provides an ideal setting to brief compliance officers, fraud investigators, and consumer protection attorneys on the underlying mechanics of our LM. Specifically, it allows technical teams to unpack how intermediate pre-training, inverse class weighting, and confidence calibration directly shape downstream routing. Because categories such as `fraud_unauthorized` carry severe financial and compliance ramifications, financial institutions cannot rely on a black-box scoring mechanism. Through deliberative sessions, participants work through realistic, anonymized edge cases—such as distinguishing between a legitimate merchant refund delay and an unauthorized third-party account takeover masquerading under ambiguous or distressed consumer phrasing.

Crucially, this method empowers institutional stakeholders to actively debate and co-determine policy-level boundaries between automated throughput and human review. Instead of developers unilaterally setting an uncertainty threshold (such as $P < 0.70$), deliberative workshops allow bank compliance officers to help define what specific margins of error are legally tolerable under consumer financial protection standards. Participants can scrutinize the practical implications of false positives versus false negatives: a false positive might unnecessarily tie up expensive investigative resources, whereas a false negative could leave an ongoing account takeover unaddressed, violating regulatory escalation timelines.

Furthermore, deliberative workshops provide a transparent space to address institutional concerns surrounding "automation bias." By discussing real-world casework alongside machine learning engineers, fraud specialists can help design decision-support criteria that prevent frontline analysts from passively rubber-stamping the LM's outputs. They can specify what supporting evidence the model should highlight in the interface—such as flagged transactional terms or anomaly timestamps—to substantiate why a complaint was triaged into a high-risk fraud bucket.

By fostering sustained, deliberative dialogue rather than brief consultation, this method bridges the communication gap between algorithmic system design and operational banking policy. Financial institutions transition from defensive, external risk managers into collaborative partners who help shape the governance architecture. The resulting operational alignment ensures that our

LM functions as a dependable, compliant triage assistant that streamlines fraud routing while preserving necessary human oversight for high-liability financial disputes.

Small Businesses' Engagement method: Focus groups are structured, small-group discussions that bring together a handful of representative participants, including small business owners, customer service staff, and e-commerce managers, to react directly to concrete material rather than abstract policy questions. This method suits small businesses because, unlike financial institutions, they typically lack dedicated compliance or legal teams and cannot commit to prolonged deliberative processes; what they need is fast, practical feedback grounded in their daily reality of handling customers.

A focus group session would present participants with anonymized complaint examples alongside the model's classification logic—explaining, in accessible terms, what phrasing patterns, keywords, and urgency cues push a complaint toward fraud_unauthorized versus a routine category. This transparency lets business owners spot where the logic misfires: for instance, a customer describing a late delivery in urgent or accusatory language, which risks being over-classified as fraud and triggering an unwarranted account freeze or chargeback penalty, versus a genuinely distressed but understated message that actually signals fraud.

This engagement directly serves the interests laid out for this stakeholder group: operational fairness, revenue stability, and reputational integrity. Businesses that clear complaint backlogs quickly avoid the reputational damage of being called out publicly for slow responses, while accurate routine-vs-fraud distinctions protect them from the lawsuits and financial loss that can follow a serious misclassification. Because small businesses deal with highly varied, informal customer language daily, their input also helps the model generalize fairly across real customer voices rather than only the more formal language financial institutions tend to use in their own records.

Focus groups also surface the stakes on both sides of the threshold debate: a false positive can unnecessarily tie up a small business's limited staff time investigating a non-issue and strain customer relationships, while a false negative — a genuine fraud case waved through as routine — exposes the business to real financial loss and legal liability. Giving business owners a voice in these tradeoffs ensures thresholds reflect what's actually survivable for a small operation, not just what's statistically optimal.

Discussions can also guard against automation bias: frontline staff should not simply rubber-stamp the model's output. Participants can help specify what the interface should surface to justify a classification — such as the flagged phrase or pattern that triggered it — so staff can quickly verify rather than blindly trust the routing.

Through this engagement, small businesses move from passively absorbing whatever classification the model assigns to actively shaping how it interprets their customers' everyday language, helping ensure the system functions as a fair, transparent triage tool rather than an opaque one.

- 6) **Plan use:** Businesses and Financial institutions interact with customers daily and deal with the nuances of customer complaints. By engaging with model builders in deliberative workshops, model developers can better incorporate these nuances of everyday business. For instance, through stakeholder deliberations, the diverse language customers use can be captured and built into the model to improve customer satisfaction.